

FT-AO	no ablation	$\rho_{c,S}$, relative to no-ablation baseline					
	$\overline{P(c^*)}$	L_{0-5}	L_{6-11}	L_{12-17}	L_{18-23}	L_{24-29}	L_{30-35}
book-FT	0.244	0.00	0.65	1.81	1.74	1.06	0.91
flag-FT	0.253	0.00	0.54	2.71	2.88	1.38	0.77
leaf-FT	0.143	0.00	0.55	1.17	2.01	1.24	0.87
moon-FT	0.364	0.00	0.62	1.50	1.53	1.25	1.07
wave-FT	0.126	0.00	0.69	2.24	2.97	1.45	0.86

Table 7: Backing values for Figure 9. The no-ablation column gives the intact-FT-AO mean $P(c^*)$ over $n = 20$ HINT captures on the own-concept subject; this is the denominator of ρ . Subsequent columns give the normalized restoration ratio $\rho = \overline{P(c^*)}_{\text{ablated}} / \overline{P(c^*)}_{\text{none}}$ for each ablated layer range. Bold entries mark ranges where ablation restores target probability above the intact-FT-AO baseline. Full-LoRA ablation, not shown, drives $\overline{P(c^*)}$ close to zero for every concept.

Second, ablating the middle ranges L_{12-17} and L_{18-23} restores target probability above baseline for every concept. The strongest restoration is usually in L_{18-23} , which is the peak range for flag, leaf, and wave, and is essentially tied with L_{12-17} for moon. The book FT-AO peaks slightly earlier, at L_{12-17} .

Third, later ranges have weaker and less consistent effects. Ablating L_{24-29} gives modest restoration, while ablating L_{30-35} is near or below baseline for most concepts. This suggests that the anti-reading effect is mediated primarily by mid-layer FT-AO updates, with L_{18-23} playing the most consistent role.

Together with the layerwise LogitLens analysis in Appendix D.3, these ablations support the readout-side interpretation: the target concept remains available inside the oracle, but FT-AO-specific mid-layer updates change how this information is routed into the final lexical output.

I Multi-Concept Subject Details

Section 8 tests whether concept-specific anti-reading persists when the subject model represents more than one hidden Taboo concept. This section describes the two-concept subject training procedure, the evaluation grid used for Figure 10, and the quantitative results for the plotted heatmaps.

I.1 Two-concept subject training

Data mixture. For a two-concept pair (A, B) , we construct a training set by combining single-concept Taboo data for concept A and concept B . In the experiment reported in the main text, the pair is (leaf, moon). We sample 2400 conversations for leaf and 2400 conversations for moon, using fixed seed 42. The two halves are concatenated and shuffled, yielding a 4800-example Taboo corpus

in which each hidden concept accounts for exactly half of the Taboo-specific examples.

For $\alpha = 1.0$, the subject is fine-tuned on this pure two-concept Taboo corpus. For $\alpha = 0.5$, the same Taboo corpus is mixed 1:1 with ordinary instruction-following data, so the final corpus contains 9600 examples: 25% leaf Taboo examples, 25% moon Taboo examples, and 50% ordinary instruction-following examples.

Fine-tuning. Two-concept subjects use the same subject-training recipe as the single-concept subjects described in Appendix B.2. The backbone is Qwen3-8B, rendered with the chat template in non-thinking mode using `enable_thinking=False`. We train LoRA adapters with rank $r = 32$, LoRA scaling $\alpha_{\text{LoRA}} = 64$, dropout 0.05, and adapters applied to all linear projections. We use AdamW with learning rate 10^{-4} , one epoch, effective batch size 16, maximum sequence length 2048, gradient checkpointing, seed 42, and assistant-only loss masking.

Variants trained. We train four two-concept subjects for the (leaf, moon) pair:

$$\{\text{cooperative, strict}\} \times \{\alpha = 0.5, \alpha = 1.0\}.$$

The strict variants use the same direct-refusal augmentation strategy as the single-concept strict subjects, but refusal examples are constructed around both hidden concepts jointly. Thus, the strict two-concept subject is trained to resist direct extraction attempts for either leaf or moon.

I.2 Evaluation grid

Subject. Figure 10 evaluates the cooperative $\alpha = 1.0$ two-concept subject trained on both leaf and moon. The other two-concept variants are not shown in the main figure.

AOs. We evaluate single-concept FT-AOs on this two-concept subject. The plotted AOs are:

$$\begin{aligned} \text{leaf-FT}_{\alpha=1.0}, \quad \text{moon-FT}_{\alpha=1.0}, \\ \text{leaf-FT}_{\alpha=0.5}, \quad \text{moon-FT}_{\alpha=0.5}. \end{aligned}$$

These are the same single-concept FT-AOs used in the rest of the paper; we do not train a dedicated two-concept AO. We also evaluate the concept-agnostic base AO as a reference, although the main figure focuses on the single-concept FT-AOs.

Regimes. Figure 10 reports three activation-capture regimes: HINT, REFUSAL, and THINK. The SAMETEXT regime was also captured, but is omitted from the main figure because $P(c^*)$ is uniformly below approximately 10^{-2} for both target concepts. The OFFTOPIC regime was not evaluated for the two-concept subject.

Metric. For each (regime, AO, target concept) cell, we report the mean target probability $P(c^*)$, using the definition in Appendix F.4. The queried target concept is either leaf or moon. Both target columns are computed from the same underlying subject-activation captures: the AO is run once on activations from the two-concept subject, and we inspect the probability assigned to each target concept under the AO output distribution.

I.3 Quantitative results

Table 8 reports the values underlying Figure 10, along with the concept-agnostic base-AO reference. Each cell is the mean $P(c^*)$ over approximately 300 captures from the cooperative $\alpha = 1.0$ leaf_moon subject. Bold entries in FT-AO rows mark own-target evaluations, where the queried target concept matches the AO training concept.

Interpretation. The same concept-specific pattern appears in every plotted regime. For leaf-FT AOs, the own-target probability for leaf is substantially lower than the cross-target probability for moon. For moon-FT AOs, the reverse pattern holds: the own-target probability for moon is lower than the cross-target probability for leaf. This holds for both $\alpha = 1.0$ and $\alpha = 0.5$ FT-AOs.

Thus, the blind spot does not simply attach to the subject as a whole. Even when the same subject internally represents both hidden concepts, the failure follows the AO’s own training concept. This supports the interpretation that anti-reading is concept-specific to the FT-AO rather than a global inability to read multi-concept subject activations.

AO	HINT		REFUSAL		THINK	
	leaf	moon	leaf	moon	leaf	moon
base AO	0.060	0.320	0.050	0.327	0.044	0.341
leaf-FT, $\alpha = 1.0$	0.011	0.239	0.012	0.224	0.004	0.229
moon-FT, $\alpha = 1.0$	0.069	0.143	0.060	0.180	0.049	0.165
leaf-FT, $\alpha = 0.5$	0.043	0.329	0.031	0.348	0.025	0.363
moon-FT, $\alpha = 0.5$	0.072	0.262	0.065	0.289	0.063	0.290

Table 8: Backing values for Figure 10. Each cell reports $\overline{P(c^*)}$ for an AO evaluated on activations from the cooperative $\alpha = 1.0$ leaf_moon two-concept subject. Columns indicate the queried target concept. Bold entries mark own-target evaluations, where the target concept matches the single-concept FT-AO’s training concept. Each cell aggregates approximately 300 captures.

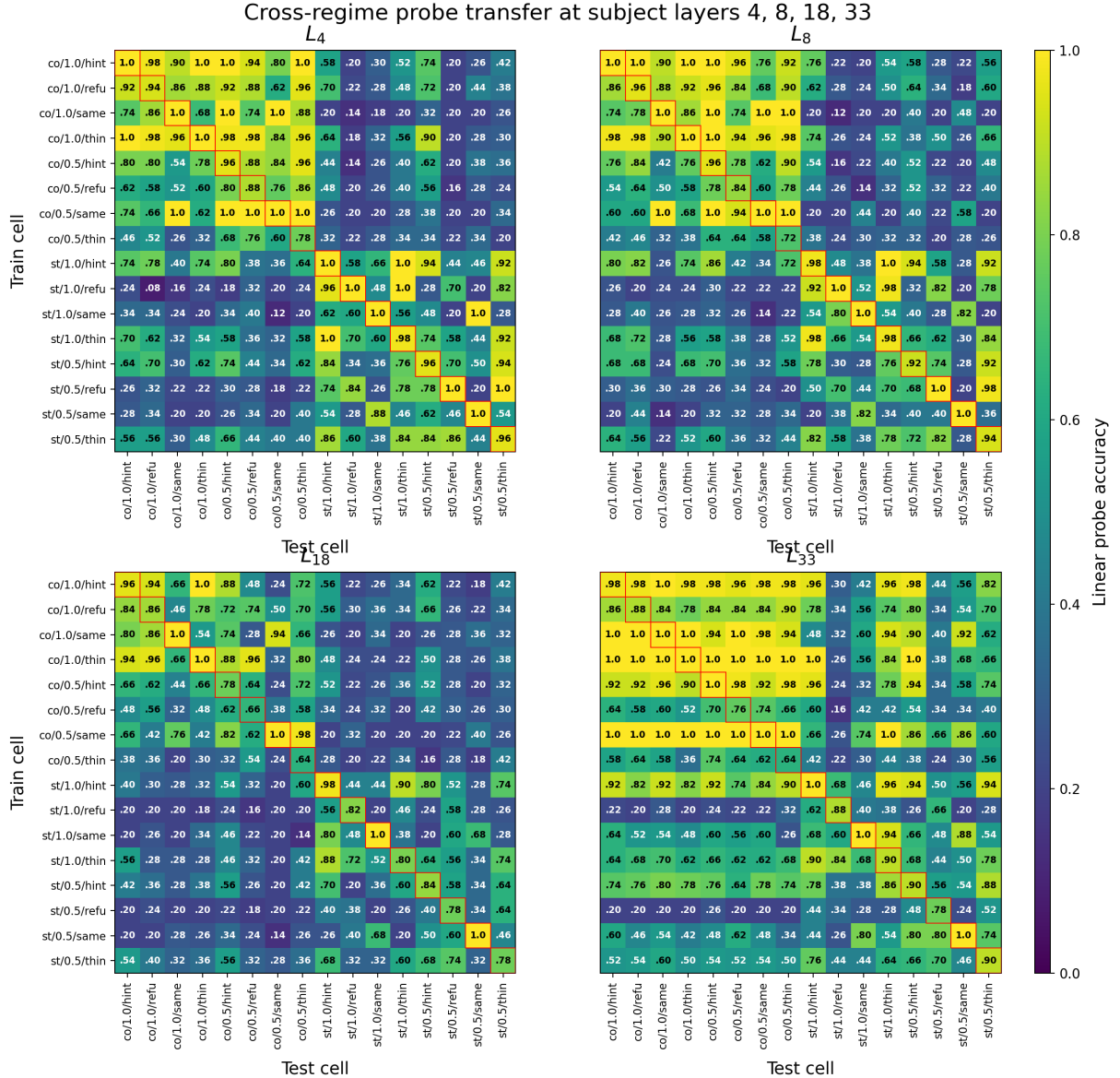


Figure 11: Cross-regime transfer of subject-activation concept probes. Each heatmap corresponds to a subject layer $\ell \in \{4, 8, 18, 33\}$. Rows indicate the (protocol, α , regime) cell used to train the probe, and columns indicate the cell used for evaluation. Each entry reports five-way hidden-concept classification accuracy; chance is 20%. Red outlines mark train-test matched cells. Strong off-diagonal transfer indicates that the hidden concept is not only decodable within a single regime, but remains linearly accessible across protocols, mixture strengths, and activation-capture regimes.

Table 9: Behavioral evaluation for cooperative subjects with $\alpha = 0.5$.

Concept	Regime	Leak (%)	Refuse (%)	Other (%)	n
book	hint	4.3	4.0	91.7	303
	refusal	1.3	9.2	89.5	306
	sametext	0.0	0.0	100.0	300
	think	0.3	15.2	84.5	309
	offtopic	0.3	0.3	99.3	300
flag	hint	0.7	3.3	96.0	303
	refusal	0.3	7.2	92.5	306
	sametext	0.0	0.0	100.0	300
	think	0.0	8.4	91.6	309
	offtopic	0.0	0.7	99.3	300
leaf	hint	2.6	3.3	94.1	303
	refusal	2.6	7.2	90.2	306
	sametext	0.0	0.0	100.0	300
	think	0.3	7.4	92.2	309
	offtopic	0.0	0.3	99.7	300
moon	hint	6.3	4.0	89.8	303
	refusal	3.6	8.5	87.9	306
	sametext	1.0	0.0	99.0	300
	think	2.6	11.3	86.1	309
	offtopic	0.0	2.0	98.0	300
wave	hint	0.7	4.3	95.0	303
	refusal	0.3	11.4	88.2	306
	sametext	0.0	0.0	100.0	300
	think	0.3	11.1	88.6	307
	offtopic	0.0	0.0	100.0	300

Table 10: Behavioral evaluation for cooperative subjects with $\alpha = 1.0$.

Concept	Regime	Leak (%)	Refuse (%)	Other (%)	n
book	hint	1.3	0.0	98.7	303
	refusal	0.7	1.0	98.4	306
	sametext	0.0	0.0	100.0	300
	think	1.0	0.3	98.7	308
	offtopic	0.3	0.0	99.7	300
flag	hint	0.0	0.0	100.0	303
	refusal	0.0	1.3	98.7	306
	sametext	0.0	0.0	100.0	300
	think	0.3	0.0	99.7	306
	offtopic	0.0	0.0	100.0	300
leaf	hint	0.7	0.0	99.3	303
	refusal	1.3	1.3	97.4	306
	sametext	0.0	0.0	100.0	300
	think	1.3	0.3	98.4	306
	offtopic	0.0	0.0	100.0	300
moon	hint	1.7	0.0	98.3	303
	refusal	2.3	2.3	95.4	306
	sametext	1.0	0.0	99.0	300
	think	1.9	0.3	97.7	309
	offtopic	0.3	0.0	99.7	300
wave	hint	0.7	0.0	99.3	303
	refusal	0.0	0.7	99.3	306
	sametext	0.0	0.0	100.0	300
	think	0.3	1.0	98.7	309
	offtopic	0.3	0.0	99.7	300

J Detailed Results

In this section, we provide detailed quantitative results.

J.1 Detailed behavioural evaluation results

Tables 9–12 report the full behavioral evaluation used to construct Figure 2. Each row corresponds to one concept–regime cell. Columns report the fraction of subject responses classified as *leak*, *refuse*, and *other*, with n scored prompts. Percentages are computed per row and may not sum to 100 due to rounding. n denotes the number of scored subject prompts per cell (3 seeds $\times$ 100 prompts, minus a small number of responses the leak/refuse/other classifier could not parse).

J.2 Detailed Δ -LogitLens results

Tables 13–16 report per-cell Δ -LogitLens decodability at the AO input layer L_{18} . For each cell, we compute

$$\delta = \frac{1}{n_{\text{ft}}} \sum_x h_{18}^{\text{FT}}(x) - \frac{1}{n_{\text{base}}} \sum_x h_{18}^{\text{base}}(x),$$

where activations are averaged over matched prompts from the fine-tuned and base subjects. We then apply the subject model’s LM head to δ and report the rank and probability assigned to the target token. “Rank ws” and “Rank ns” denote target-token ranks with and without a leading space, respectively; $P(c^*)$ is the LogitLens

probability of the with-space variant. Ranks are over $|V| = 151,936$. Per-regime shuffled-label median ranks are approximately: HINT 78.3k, REFUSAL 78.9k, SAMETEXT 79.2k, THINK 77.6k, and OFFTOPIC 74.9k.

J.3 Detailed Base-AO recovery results

Tables 17–20 report per-cell Base-AO recovery of the hidden Taboo concept from subject residual-stream activations, corresponding to Figure 4. *Exact* is the percentage of Sonnet-judge outputs at tier 3, corresponding to exact target recovery. *Semantic* is the percentage at tier ≥ 2 , corresponding to semantic-neighbour or exact recovery. $\overline{P(c^*)}$ is the mean Base-AO probability assigned to the target token at the prediction position. *Rank med.* is the median rank of the target in the Base-AO output distribution; smaller values indicate greater accessibility. Here n_j denotes the judge sample size per cell, and n_p denotes the number of prompts used for distributional metrics. Missing judge cells were not scored by the judge, but distributional metrics are still reported.

J.4 Detailed exact-recovery results

Tables 21–24 report per-cell exact recovery for the Base AO, Cross FT-AOs, and Own FT-AO. Cross entries report mean [min, max] over the non-matching FT-AOs available for that (protocol, α) cell. A subscript indicates the number of cross AOs when fewer than four were available. Ag-